

SmileCare Dental System

Comprehensive Technical Architecture, API Specification & Data Model

Document Version:	1.0.0 (Production / Enterprise Edition)
System Core:	FastAPI (Python) + Next.js 16 (React / App Router)
Database Engines:	PostgreSQL (Production) / SQLite (Local & Development)
Authentication:	JWT Dual-Token Security (Staff & Patient Scopes)
Role Coverage:	Admin, Receptionist, Doctor, Lab Technician, Accountant, Patient
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Executive Summary

The **SmileCare Dental Management System** is an end-to-end, multi-tenant capable clinic management platform designed specifically for multi-specialty dental practices. The architecture bridges front-office reception, interactive 32-tooth clinical treatment charting, dental lab order tracking pipelines, financial accounting, and a patient self-service portal. Built on modern FastAPI (Python 3.11+) and Next.js 16 (App Router), SmileCare delivers real-time appointment scheduling, automated WhatsApp/SMS notifications via Twilio, IMAP automated supplier lab email processing, and insurance claim auditing.

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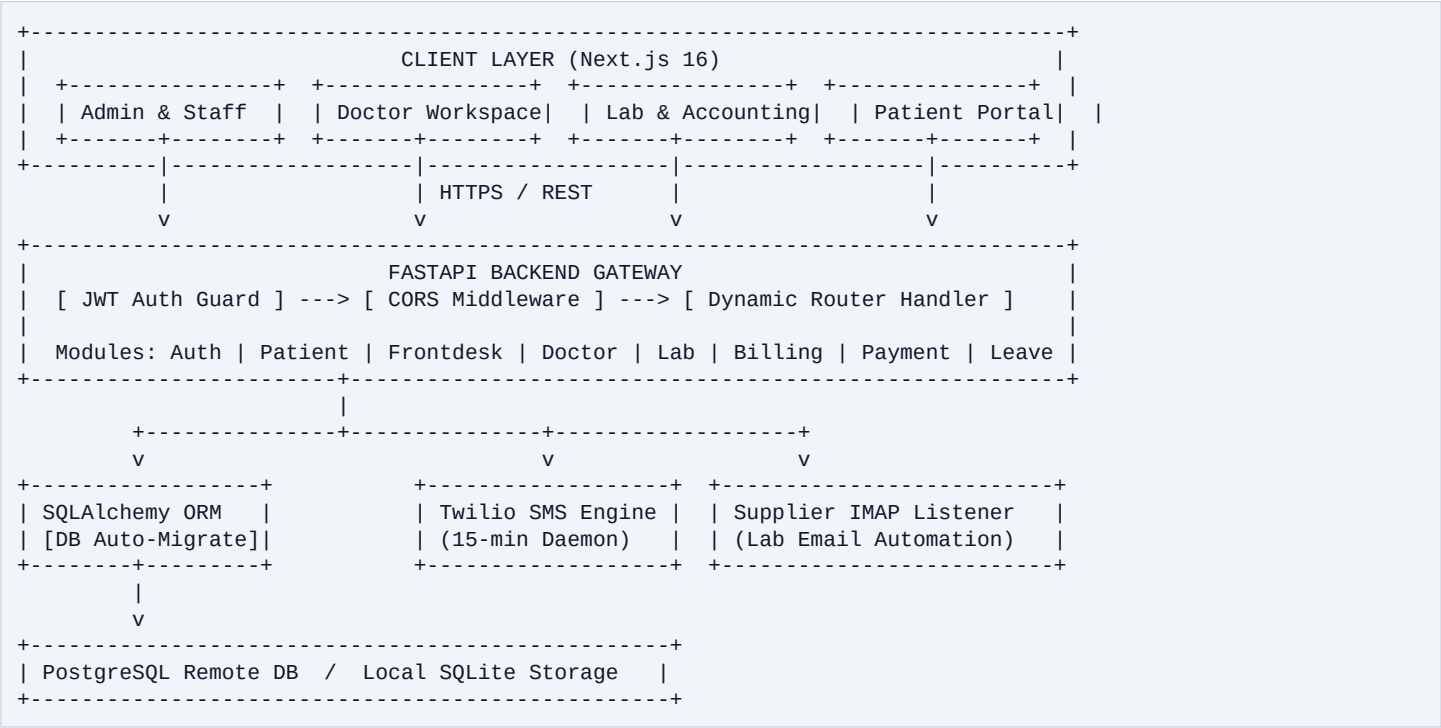
Chapter 1: System Architecture & Infrastructure Stack

SmileCare utilizes a decoupled Client-Server architecture. The frontend application is constructed using Next.js 16 (React 19 with App Router) and communicates asynchronously via REST APIs with a lightweight, high-performance FastAPI (Python) backend service. Database persistence supports both cloud PostgreSQL (with SSL connection pooling) and local SQLite for instant offline development.

Technology Stack Matrix

Component Layer	Technology / Framework	Key Responsibilities
Frontend Client	Next.js 16, React 19, Tailwind CSS, Lucide Icons	Role-based dashboards, responsive UI, 32-tooth interactive charting, Axios client.
Backend REST API	FastAPI 0.100+, Pydantic v2, Uvicorn, Starlette	13 modular domain routers, OpenAPI docs generation, request validation, CORS middleware.
Database ORM	SQLAlchemy 2.0 ORM	Entity modeling, dynamic table auto-migrations, session transaction management.
Database Engines	PostgreSQL 15+ / SQLite 3	Relational data store. Automatic fallback to SQLite if PostgreSQL URL fails.
Authentication	PyJWT, Passlib, bcrypt	Dual-token JWT creation (Staff & Patient tokens), secure password hashing.
Notification Engine	Twilio REST API, Python Threading	Automated WhatsApp & SMS appointment reminders, 15-minute background sweep.
Email Processing	IMAPlib, Python email parser	Background listener for supplier lab update emails and attachment ingestion.

System Architecture Topology Diagram



Chapter 2: Security Architecture & RBAC Matrix

SmileCare enforces a strict Role-Based Access Control (RBAC) security model. The authentication system employs dual token generation: Staff users receive role-scoped JWT tokens while Patients receive dedicated patient-token credentials upon login or registration. Password credentials are encrypted using standard bcrypt algorithms with salted rounds.

Role-Based Permission Access Matrix

Domain / Resource	Admin	Reception	Doctor	Lab Tech	Accountant	Patient
User Management	Full	None	None	None	None	None
Patient Registry	Full	Full	Read	None	Read	Self Only
Appointments & Queue	Full	Full	Assigned	None	Read	Self Only
Tooth Charting & SOAP	Read	None	Full	None	None	Read Only
Lab Orders Pipeline	Full	Pickup	Create/Rework	Full Exec	Verify/Pay	None
Billing & Payments	Full	Collect	Request	Invoices	Full Exec	Self Invoices
Procedure Catalog	Full	Read	Read	Read	Read	Read
Leave Approval	Approve	Apply	Apply	Apply	Apply	None
Developer Support	Full	Create	Create	Create	Create	None

Chapter 3: Database Architecture & Entity-Relationship Schema

The SmileCare database consists of **45 relational tables** categorized into 8 core business functional domains. The ORM implements dynamic runtime column additions via ALTER TABLE migrations, guaranteeing zero-downtime schema evolution for PostgreSQL and SQLite environments.

1. Core Identity & User Governance

Table Name	Description & Entity Role
users	Primary user model storing authentication credentials, role arrays (Admin, Doctor, Receptionist, etc.), and status flags.
staff_profiles	Extended staff metadata including phone numbers, date of birth, emergency contacts, and home address.
admin_settings	Key-value pair system configuration settings reserved for administrator control.

2. Patient Management & Portal Domain

Table Name	Description & Entity Role
patients	Patient master demographic records, assigned unique access tokens (e.g. PAT-XXXX), DOB, gender, blood group, contact.
patient_consent	Digital informed consent request forms, patient signatures, and PDF document storage paths.
patient_prescriptions	Medication prescription records containing structured JSON arrays of dosages and directions.
patient_notifications	In-app notifications pushed to patient portal dashboards.
doctor_feedbacks	Patient rating and written feedback submissions for treating doctors.
patient_clinical_notes	SOAP clinical visit progress notes created by doctors per patient token.
medicine_dispenses	Pharmacy medicine fulfillment and collection records.

3. Clinical Workspace & Interactive Tooth Charting

Table Name	Description & Entity Role
doctors	Doctor profile registry mapping user IDs to clinical specialties and room assignments.
doctor_shifts	Doctor duty shifts and availability schedules by day of week.
clinical_encounters	Encounter logs tracking patient workspace visits.
dental_charts	Master dental chart containers bound to patient tokens.
teeth	32-tooth individual state entries tracking tooth status (present, missing, impacted, crown, etc.).
tooth_surfaces	Per-tooth 5-surface breakdown (Mesial, Distal, Occlusal, Buccal, Lingual).
clinical_findings	Clinical conditions, restorations, and pathology notes mapped to specific teeth.

4. Frontdesk, Scheduling & Communication

Table Name	Description & Entity Role
appointments	Appointment bookings linked to patients, doctors, status (Scheduled, Checked-In, Completed, Cancelled), fees.
communication_logs	Twilio automated SMS/WhatsApp message dispatch logs with delivery status.
fitting_appointments	Prosthetic lab case fitting trial appointments scheduled with patients.
shift_reconciliations	End-of-day receptionist cash & card payment handover reconciliations.

5. Dental Laboratory Pipeline & Vendor Ecosystem

Table Name	Description & Entity Role
lab_orders	Master prosthetic & pathology order cases with 75 schema fields covering rework, CAD specs, stages, and tracking.
lab_vendors	External dental manufacturing labs with contact details, TAT estimates, and ratings.
lab_vendor_pricings	Custom price agreements per restoration item per vendor.
lab_inventory_items	Lab consumables stock tracking with reorder thresholds.
lab_restock_requests	Inventory restock purchase orders submitted to lab management.
prosthetic_case_details	Prosthetic specs: margin design, impression type, tooth quadrant, shade.
pathology_case_details	Pathology biopsy case specs, test types, and sample details.
lab_pricing_catalog	Master admin lab price catalog with material tiers, clinic markup %, patient pricing.
lab_order_comments	Internal discussion comments per lab order case.
lab_audit_trails	Full status progression audit log per lab order.
lab_notifications	Lab-specific task alerts for technicians.
unmatched_lab_emails	IMAP received supplier emails unmatched to active order numbers for admin review.
supplier_payables	Accounts payable entries for external lab vendor invoices.

6. Billing, Ledger & Financial Accounting

Table Name	Description & Entity Role
billing_requests	Doctor-submitted procedure billing requests sent to accounting.
invoices	Official patient invoices generated from procedures and lab fees.
payments	Payment transactions recorded against invoices (Cash, Card, UPI, Insurance).
expenses	Clinic operational overhead expense entries.
insurance_claims	Insurance pre-authorization and claim audit submission records.
consultation_payments	Frontdesk consultation fee payment receipts.
transactions	Master financial audit ledger entries.

7. Treatment Planning & Clinical Procedures

Table Name	Description & Entity Role
procedures	Standard dental procedure catalog (ADA codes, base costs, duration).
treatment_plans	Multi-stage patient treatment plan containers.
treatment_plan_steps	Individual step sequence execution within treatment plans.

8. Operations & Human Resources

Table Name	Description & Entity Role
leave_requests	Staff leave application forms with admin approval status.
clinic_settings	Global clinic operational settings (consultation fees, clinic timings).
complaints	Developer Support Portal bug reports and staff issue submissions.
complaint_logs	Status change history logs for submitted support tickets.

Chapter 4: FastAPI Backend REST API Specification

The SmileCare backend API comprises **13 modular routers** containing over **100 REST endpoints**. All protected endpoints validate Bearer tokens using FastAPI dependency injection (`get_current_user`).

Auth Router (`/auth`)

Method	Endpoint Path	Description & Function
POST	<code>/auth/login</code>	Staff & patient login. Returns JWT access token & user profile.
POST	<code>/auth/register-staff</code>	Admin creation of new staff users (Doctor, Receptionist, Lab Tech, etc.).
GET	<code>/auth/me</code>	Fetch currently authenticated user details.
PUT	<code>/auth/me/profile</code>	Update logged-in user profile details.
POST	<code>/auth/me/avatar</code>	Upload user profile avatar image.
GET	<code>/auth/users</code>	Admin endpoint listing all staff users with role filtering.

Frontdesk & Appointments Router (`/frontdesk`)

Method	Endpoint Path	Description & Function
GET	<code>/frontdesk/queue/today</code>	Fetch today's patient arrival queue board.
POST	<code>/frontdesk/appointments</code>	Schedule a new patient appointment.
GET	<code>/frontdesk/appointments</code>	Query appointments by date, doctor, or status.
PUT	<code>/frontdesk/appointments/{id}/checkin</code>	Update patient status to Checked-In.
POST	<code>/frontdesk/reminders/send-manual</code>	Trigger instant SMS/WhatsApp reminder to patient via Twilio.
GET	<code>/frontdesk/support/tickets</code>	View developer support portal tickets.

Doctor Workspace Router (`/doctor`)

Method	Endpoint Path	Description & Function
GET	<code>/doctor/today-patients</code>	Retrieve today's assigned appointment queue for doctor room.
POST	<code>/doctor/prescriptions</code>	Issue structured medicine prescription to patient.
POST	<code>/doctor/referrals</code>	Create outbound specialist referral for patient.
GET	<code>/doctor/analytics</code>	Fetch doctor performance KPIs (patients seen, average time, rating).

Dental Lab Management Router (`/lab`)

Method	Endpoint Path	Description & Function
POST	/lab/orders	Submit new prosthetic or pathology lab order.
GET	/lab/orders	Filter orders by status stage (New, In Progress, Quality Control, Dispatch).
POST	/lab/orders/{id}/rework	Flag order for rework with detailed reason and photos.
POST	/lab/orders/{id}/send-to-vendor	Dispatch lab order to external supplier lab.
POST	/lab/inbound-email	Webhook endpoint for incoming supplier lab update emails.
GET	/lab/inventory	View lab inventory consumables stock levels.

Patient Portal Router (`/patient`)

Method	Endpoint Path	Description & Function
POST	/patient/register	Public patient self-registration endpoint.
GET	/patient/profile	Fetch patient portal profile and medical history.
GET	/patient/prescriptions	Retrieve patient's past prescriptions.
POST	/patient/consents/{id}/sign	Record digital patient signature on consent form.

Billing & Payment Routers (`/billing`, `/payment`)

Method	Endpoint Path	Description & Function
POST	/billing/requests	Doctor submits completed procedure billing request.
POST	/payment/create-order	Initiate payment gateway transaction.
POST	/payment/verify	Verify payment signature and generate paid invoice.
POST	/payment/shift-handover	Receptionist submits end-of-day cash handover reconciliation.

Chapter 5: Automated Background Services & Integration Engines

SmileCare incorporates two continuous background worker engines that run as daemon threads alongside the FastAPI HTTP server:

1. Automated Appointment Reminder Engine

A background thread launches at application startup (`main.py:on_startup``) and executes a **15-minute sweep cycle**. It scans upcoming appointments scheduled for the next 24 hours, formats customized WhatsApp and SMS text messages, and dispatches them via the Twilio Messaging API (`modules/frontdesk/reminder_service.py``). Dispatch timestamps are logged to `communication_logs`` to prevent duplicate reminder dispatches.

2. IMAP Supplier Lab Email Listener

External dental labs communicate order completions, delay alerts, or rework queries via email. SmileCare runs an IMAP email listener (`modules/lab/email_listener.py``) that polls the designated lab inbox. Incoming emails are parsed for order tracking numbers (e.g. `LAB-2026-XXXX``). If matched, order statuses update automatically and attached PDF/image files are extracted to static file storage. Unmatched emails route to `unmatched_lab_emails`` for admin manual mapping.

Chapter 6: Frontend Application & Client UI Architecture

The frontend is built using Next.js 16 (App Router) structured into 6 primary role-based dashboard suites, spanning **~90+ distinct UI screens**.

- **Admin Dashboard Suite:** User CRUD management, role assignments, billable procedure catalog, global lab pricing markup controls, leave request approvals, staff complaint box, system audit logs.
- **Receptionist Dashboard Suite:** Live patient queue board, appointment calendar, patient check-in workflow, Twilio SMS message composer, Developer Support Portal (bug logging by category with status tracking).
- **Doctor Clinical Workspace:** Patient summary banner, interactive 32-tooth charting canvas (per-tooth findings, crowns, fillings, extractions), SOAP clinical notes editor, lab order submission form, prescriptions, outbound referrals.
- **Lab Technician Suite:** Kanban case pipeline (New -> In Progress -> Quality Control -> Dispatch), rework management with tooth margin specs, CAD design file upload, courier tracking, vendor invoices, inventory restock.
- **Accountant Dashboard Suite:** Revenue analytics, invoice generation, payment receipts, outstanding patient dues ledger, expense tracking, staff payroll processing, insurance claim verification workflow.
- **Patient Self-Service Portal:** Appointment calendar, online check-in, prescription history, digital consent form signing, invoice receipts, personalized dental care tips.

Chapter 7: Business Workflows & Lifecycle Logic

End-to-End Clinical & Lab Order Workflow

1. **Patient Check-In:** Receptionist checks in arriving patient. Patient appears on Doctor Room Queue.
2. **Clinical Examination:** Doctor opens Clinical Workspace, inspects 32-tooth chart, adds findings, and enters SOAP notes.
3. **Treatment Plan & Procedure:** Doctor adds procedure (e.g. Zirconia Crown on Tooth #14) and generates Lab Order.
4. **Lab Order Dispatch:** Order enters Lab Technician Kanban ('New Cases'). Technician assigns vendor and dispatches.
5. **QC & Fitting:** Lab receives completed crown, conducts QC checklist, and schedules Fitting Appointment.
6. **Rework Loop (If needed):** If shade or fit is incorrect, Doctor clicks 'Request Rework'. System creates linked rework order.
7. **Billing & Handover:** Accountant verifies invoice. Patient pays via Cash/UPI/Card. Shift handover reconciled.

Chapter 8: Deployment, Migration Strategy & Operations Manual

Environment Setup & Execution Commands

```
# 1. Backend Server Setup
cd backend
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
uvicorn main:app --reload --port 8000

# 2. Frontend Application Setup
cd frontend
npm install
npm run dev -- -p 3000

# 3. Seed Specialty Data Scripts
python seed_general_dentistry.py
python seed_oral_surgery.py
python seed_prosthodontics.py
```

Verification & System Health Checklist

- Verify API docs accessible at <http://localhost:8000/docs>.
- Confirm database table creation and dynamic migrations execute cleanly on boot.
- Check background worker threads start successfully without socket exceptions.
- Validate Next.js 16 frontend page routing across all 6 dashboard roles.